

Sergio Rivera Bustamante

ML Systems Engineer | High-Performance AI Infrastructure Viña del Mar, Chile |
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Professional Summary

Systems-oriented ML Engineer bridging the gap between Python experimentation and bare-metal execution. Specialized in high-performance infrastructure, SRE principles, and Rust (PyO3) integration to overcome CPython's memory and latency limitations. Proven track record in designing autonomous SRE agents and optimizing CV pipelines for clinical-grade numerical integrity and cost-efficiency.

Technical Skills

- **Systems & Low-Level:** Rust (Tokio, Axum, PyO3), SIMD (AVX2/SSE), Zero-Copy processing, CPython heap optimization.
- **AI/ML Engineering:** PyTorch (Inference optimization), LLM Orchestration, Autonomous AI Agents, RAG, Transfer Learning.
- **Infrastructure & SRE:** Kubernetes, Prometheus/Alertmanager, CI/CD, SSH/K8s automation, Networking (CCNP/Juniper background).
- **Tools & MLOps:** ZenML, Weights & Biases (W&B), BentoML, FastAPI, Streamlit, Docker, Git.
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Professional Experience

Independent R&D (ML Systems Engineer) | May 2026 – Present

- **AI SRE Platform:** Engineered a hybrid autonomous agent (Python + Rust) for real-time cybersecurity incident response.
 - Developed a native Rust module for ultra-fast alert validation (<5ms p99 latency) and dynamic blocking using async Axum/Tokio.
 - Implemented a strict command validation harness (allow/deny-lists) to secure critical infra, achieving 100% interception of destructive commands.
 - Designed ingestion capacity for >10,000 alerts/sec with zero-latency overhead.
- **CV Pipeline Optimization:** Migrated legacy PIL → PyTorch pipelines to a Rust/PyO3 architecture for clinical diagnostic support.
 - Eliminated buffer duplication using zero-copy techniques and SIMD-accelerated resizing (fast_image_resize).

- Achieved 87% reduction in memory (RSS) and 1.6x latency speedup (2.61s → 1.62s per batch).
- Certificated numerical equivalence for ResNet50 (MSE 3.37e-05), ensuring diagnostic integrity in clinical environments.

Data Scientist (Internships & Apprenticeships) | Aug 2024 – Oct 2025

- **Electricity Demand Forecasting:** Implemented an XGBoost model with robust feature engineering to predict power demand, achieving >98% accuracy (1.54% MAPE). Delivered via interactive Streamlit dashboard.
- **MLOps Orchestration:** Built end-to-end churn classification pipelines using ZenML and BentoML, reducing deployment friction from days to minutes.
- **Medical Imaging Prototype:** Developed a binary classifier (ResNet50) for MRI scans, providing a data-driven framework for deep learning ecosystem selection (PyTorch vs. TensorFlow).

First Line Engineer, Bridgestone | Jan 2023 – Jan 2024

- Managed critical customer incidents and infrastructure support, developing the core SRE/Networking foundation and customer-facing communication skills. (Handled 1,000+ customer interactions).

Education

Federico Santa María Technical University | Viña del Mar, Chile

University Technician, Telecommunications and Networks

- Key foundation for networking, concurrency, and high-availability systems.

Languages

- **Spanish:** Native
- **English:** Advanced (Professional Working Proficiency)
- **German:** Intermediate